



Assessing Proximity and Access to Healthcare Services in Chicago

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Proximity and Access to Healthcare

Healthcare is a fundamental human right that everyone should have access to. It is one thing to be close in proximity to a health care location, and another to have access to it.

Social determinants of health include: aspects of the social environment and the physical, built environment. They include economic stability, education, access to transportation systems, access to healthcare. This is why I looked at public transportation data to assess access to healthcare.



The City of Chicago

Within cities, there is a lower demand for car ownership since their public transportation systems are ranked highly.

Chicago's transportation system ranked 6th in the country, which was calculated based on how far a person would have to travel from their home to the transit stop, the frequency, and the variety of public transit. It was considered within the "good" category based on the Walk Score calculated.

For reference, New York City, ranked number 1, scored just over a 84, while Chicago's was just under 65.

Data Used

All data was obtained from the City of Chicago Official Website and the United States Census Bureau. The data sets included:

- Primary Health Care Service Locations
- Hospital data on name and location
- School name and location data
- Number and location of bus stops
- Unemployment data for community areas in Chicago

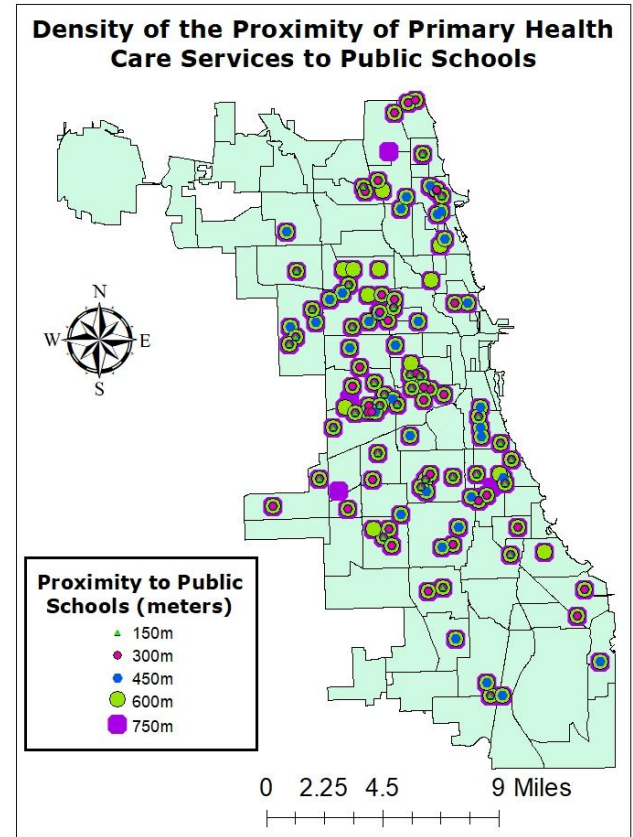
Proximity of Primary Healthcare Locations to Schools

Proximity of healthcare service locations to schools was measured by creating different layers for the different distances from schools.

Here is a map of the findings.

The density of healthcare services was greatest among the distance of 150 meters from school locations, which shows an adequate number of healthcare services close to public schools.

The figure to the right depicts this data, within the specified parameters.

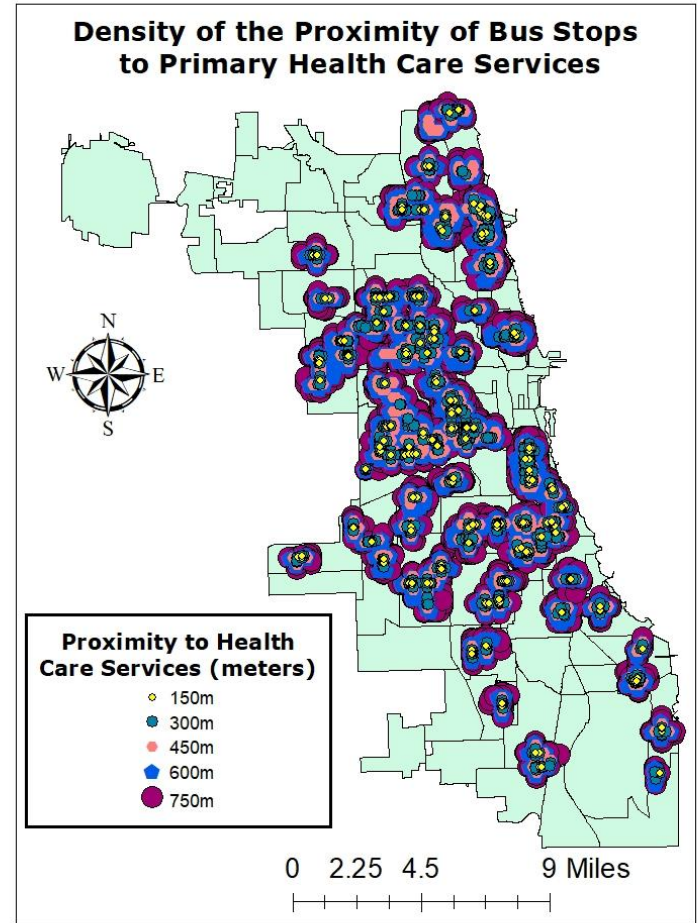


Primary Health Locations and Bus Stops

Different layers were created for the widened distance from bus stops and the densities were found.

I found that the highest density of bus stops was within the 750 meter range.

The second highest density was within the 150 meters from health care service range, which shows that the number of bus routes close to healthcare locations is adequate although it can still be improved.



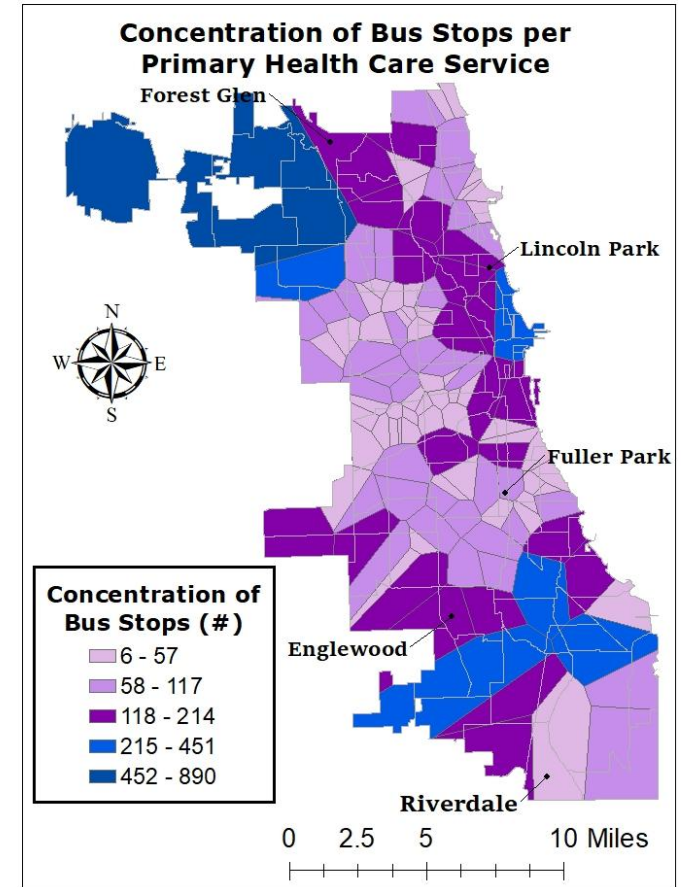
Concentration of Bus Stops

I created Thiessen polygons for each healthcare service and found how many bus stops were within that polygon.

In the figure, the Thiessen polygons are divided into different sections by the concentration of bus stops in the area.

Specific community areas are labeled to show a pattern.

The lower the unemployment rate and the higher the poverty rate, the less amount of bus stops are within the area.

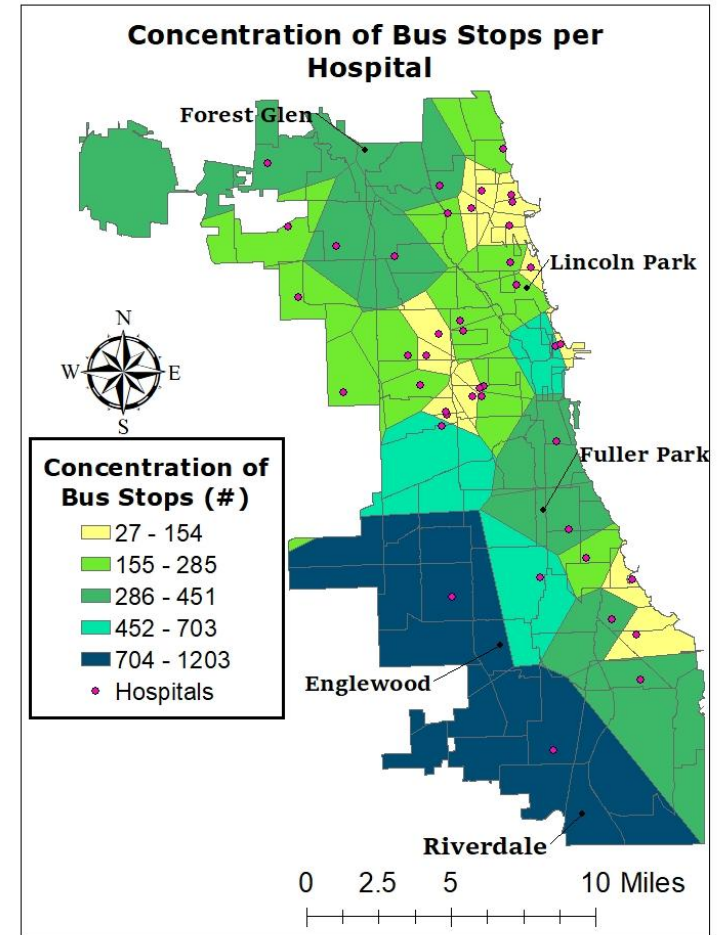


Concentration of Bus Stops pt. 2

The highest density found was within the 750 meter range, meaning that the majority of bus stops were farther away from hospitals.

The number of hospitals is concentrated within the North Side of Chicago, while the South side, particularly the Far Southwest side does not have many hospitals.

Englewood and Riverdale each only have one hospital in their community area.



Is there a pattern here?

I had predicted that there would be a lower concentration of bus stops within the neighborhoods with higher rates of unemployment, but this was not the case.

The Thiessen polygons were larger within the areas with higher unemployment because the number of hospitals were sparse.

There are many more hospitals close to neighborhoods like Fuller Park and Lincoln Park that would be easier to travel to by bus than from Riverdale to the nearest hospital.

Looking Forward

People living in areas with higher concentrations of health care are more likely to report better health. Thus, to further my analysis, I would like to look closely at data by neighborhood, income, and race as Chicago is extremely segregated and has a big inequality gap among its residents.

Also, data on the funding and quality of hospitals would be noteworthy.